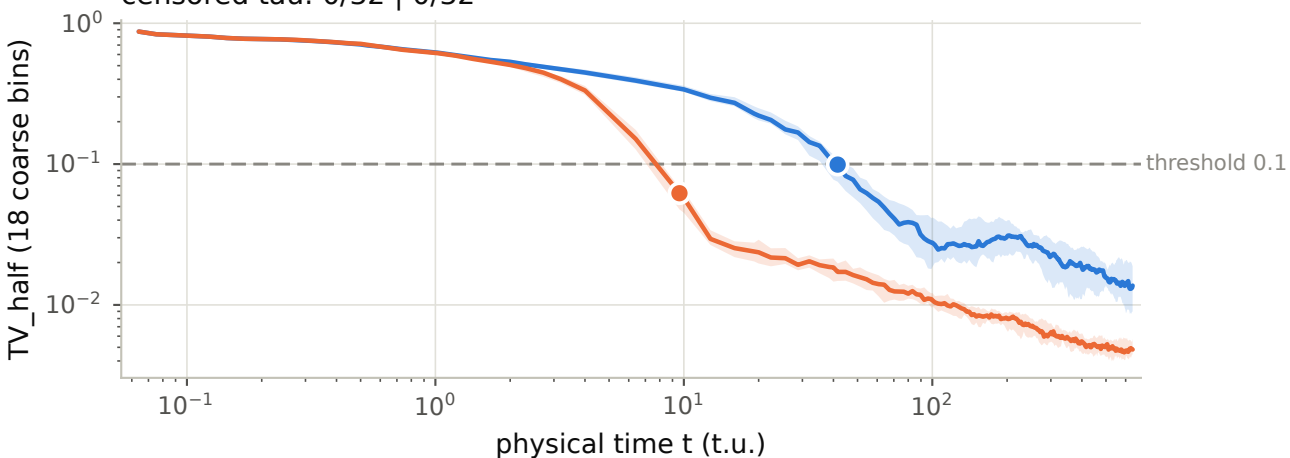


Gateway family matched_free_energy/alpha0.5 · N = 128 · T_N = 640 t.u.

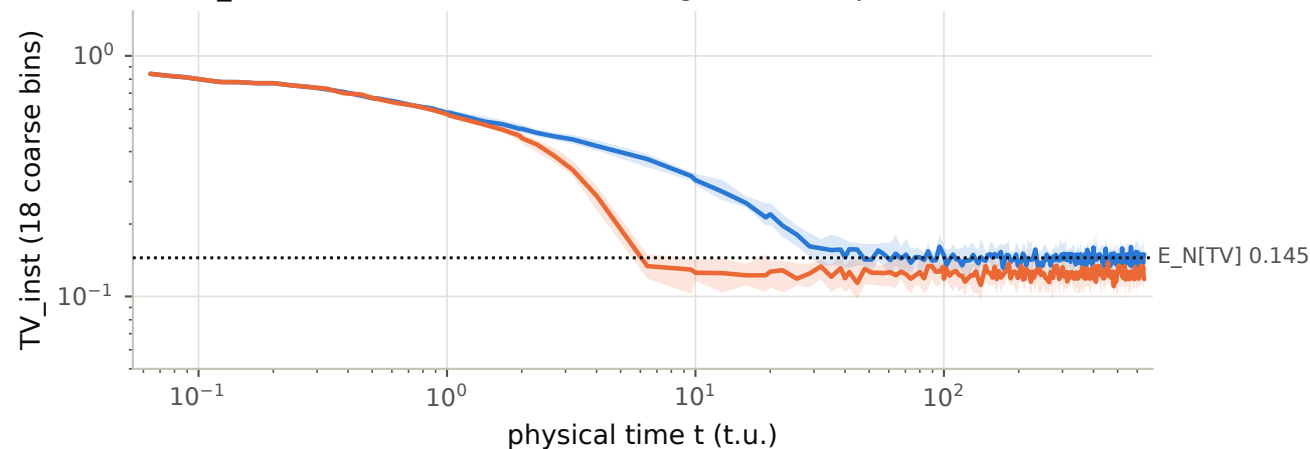
n_steps = 25,600,000 · h = 2.5e-05 · B = N n_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Marginal convergence: TV_half and TV_inst with the finite-N floor vs physical time

(a) TV_half: visitation over the trailing half $C_{all}(t) - C_{all}(\approx t/2)$
censored tau: 0/32 | 0/32



(b) TV_inst: instantaneous walker histogram (descriptive)



- ABF (n = 32)
- ABF+FR (n = 32)
- interquartile range
- O — TV_half threshold 0.1 (marker: persistent crossing of the median curve)
- finite-N floor $E_N[TV] = 0.1447$ (exact, multinomial(N, uniform))

Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

TV_half uses the saved step nearest $t/2$ (or the run start) as the window's lower edge (eqb_metrics.tv_half); tau for the marginal is defined on TV_half. TV_inst is never used for tau: at small N it is dominated by discreteness (17/18 at $N = 1$).

eqb_plot_config/2 · config alpha0.5.json · 2026-10-10 14:20 UTC